

Code: I7AZQ

The velocity of a particle travelling along the x axis is given by $v = 2 - \frac{10}{t+2} \text{ ms}^{-1}$, where t is the time in seconds .

a) When is the particle stationary? 1

.....

.....

.....

.....

.....

b) What happens to the velocity as $t \rightarrow \infty$? 1

.....

.....

.....

.....

.....

c) Find the acceleration when the particle is stationary. 2

.....

.....

.....

.....

- d) Find the distance travelled in the first 5 seconds, in exact form. 3

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Code: 3BXAO

- (i) Show that $1 + e^{-x} = \frac{e^x + 1}{e^x}$. 1

- (ii) The velocity v of a particle moving along the x -axis is given by 3

$$\frac{dx}{dt} = 1 + e^{-x} \text{ where } x \text{ is the displacement of the particle from the origin}$$

in metres. Initially the particle is at the origin.

Find the time taken by the particle to reach a velocity of $1\frac{1}{2} \text{ ms}^{-1}$.

Code: GAUFU

- b) The velocity ' v ' of a particle along the ' x ' axis is given by $v = \sqrt{25 - x^2}$ where ' x ' is the displacement of the particle from O. Initially the particle is 5cm to the right of O.

Find an expression for ' x ' in terms of ' t ' (4)

Code: TTL17

8. Calculus, 2ADV C3 2011 HSC 7b

The velocity of a particle moving along the x -axis is given by

$$v = 8 - 8e^{-2t},$$

where t is the time in seconds and x is the displacement in metres.

- Show that the particle is initially at rest. (1 mark)
- Show that the acceleration of the particle is always positive. (1 mark)
- Explain why the particle is moving in the positive direction for all $t > 0$. (2 marks)
- As $t \rightarrow \infty$, the velocity of the particle approaches a constant.

Find the value of this constant. (1 mark)

- Sketch the graph of the particle's velocity as a function of time. (2 marks)

Code: KWA30

- a) A particle starts from the origin and moves in a straight line with a velocity given by

$$v = \frac{3t}{5+t^2} \text{ metres per second, where } t \text{ is in seconds.}$$

- Find an expression for the acceleration. Simplify your answer. 2
- Describe the motion of the particle initially. In your answer, make reference to the velocity and to the acceleration of the particle. 2

Code: PBVBH

- a) Car A and car B are travelling along a straight level road at constant speeds V_A and V_B respectively. Car A is behind car B, but is travelling faster.

When car A is exactly D metres behind car B, car A applies its brakes, producing a constant deceleration of $k \text{ m/s}^2$.

- Using calculus, find the speed of car A after it has travelled a distance x metres under braking. 2
- Prove that the cars will collide if $V_A - V_B > \sqrt{2kD}$. 4

Code: MA24C

The motion of a particle moving in a straight line is described by $v = \frac{6}{t+3}$, where v is the velocity in metres/second and t is the time in seconds. Initially the particle is at the origin.

- (i) What is the acceleration of the particle 2 seconds after the start of the motion? 2
- (ii) Find the displacement, x , as a function of time. 2
- (iii) Briefly describe the motion of the particle. 2

Code: MA58W

- c) A certain particle moves along the x axis according to $t = 2x^2 - 5x + 3$ where x is measured in metres and t in seconds.

Initially the particle is 1.5m to the right of O and moving away from O .

- i) Prove that the velocity, $v \text{ ms}^{-1}$, is given by $v = \frac{1}{4x - 5}$. 2
- ii) Find an expression for the acceleration in terms of x . 2
- iii) Find the velocity of the particle when $t = 6$ seconds. 3

Code: QGSW7

A particle starts from rest at O and moves along the x axis so that its acceleration after t secs is $(24t - 12t^2) \text{ m/sec}$.

- i) Find when the particle again returns to O and its velocity at that time. 2
- ii) What is the farthest that the particle travels from O during this interval. 1

Code: C6JN9

- c) (i) A particle is travelling in a straight line. Its displacement from the origin is $x \text{ m/s}$ at time t seconds. 2

If $x = \sqrt{3} \cos 2t - \sin 2t$, express x in the form of $R \cos(2t + \alpha)$, where $R > 0$ and $0 \leq \alpha \leq 2\pi$.

- (ii) Find the maximum speed of the particle and the time when it first occurs. 2

Code: D83JS

c) A particle moves along a straight line about a fixed point O so that its

acceleration, $a \text{ ms}^{-2}$, at time t seconds is given by $a = 4 \cos\left(2t + \frac{\pi}{6}\right)$.

Initially the particle is moving to the right with a velocity of 1 ms^{-1} from a

position $\frac{\sqrt{3}}{2}$ metres to the left of O .

- | | | |
|-------|---|---|
| (i) | Find an expression for the velocity of the particle after t seconds. | 2 |
| (ii) | Find an expression for the position of the particle after t seconds. | 2 |
| (iii) | Show that the particle changes directions when $t = \frac{5\pi}{12}$ seconds. | 1 |
| (iv) | At what time does the particle return to its initial position for the first time? | 1 |

Code: XKT63

A particle is moving on the x -axis. Its velocity, v metres per second, at time t seconds is given by $v = 3 - 6 \cos t$.

- | | | |
|-------|---|---|
| (i) | What is the maximum velocity of the particle? | 1 |
| (ii) | When does the particle first come to rest? | 1 |
| (iii) | Sketch the graph of v as a function of t for $0 \leq t \leq 2\pi$. | 1 |
| (iv) | Find the total distance travelled by the particle between $t = 0$ and $t = \pi$. | 3 |